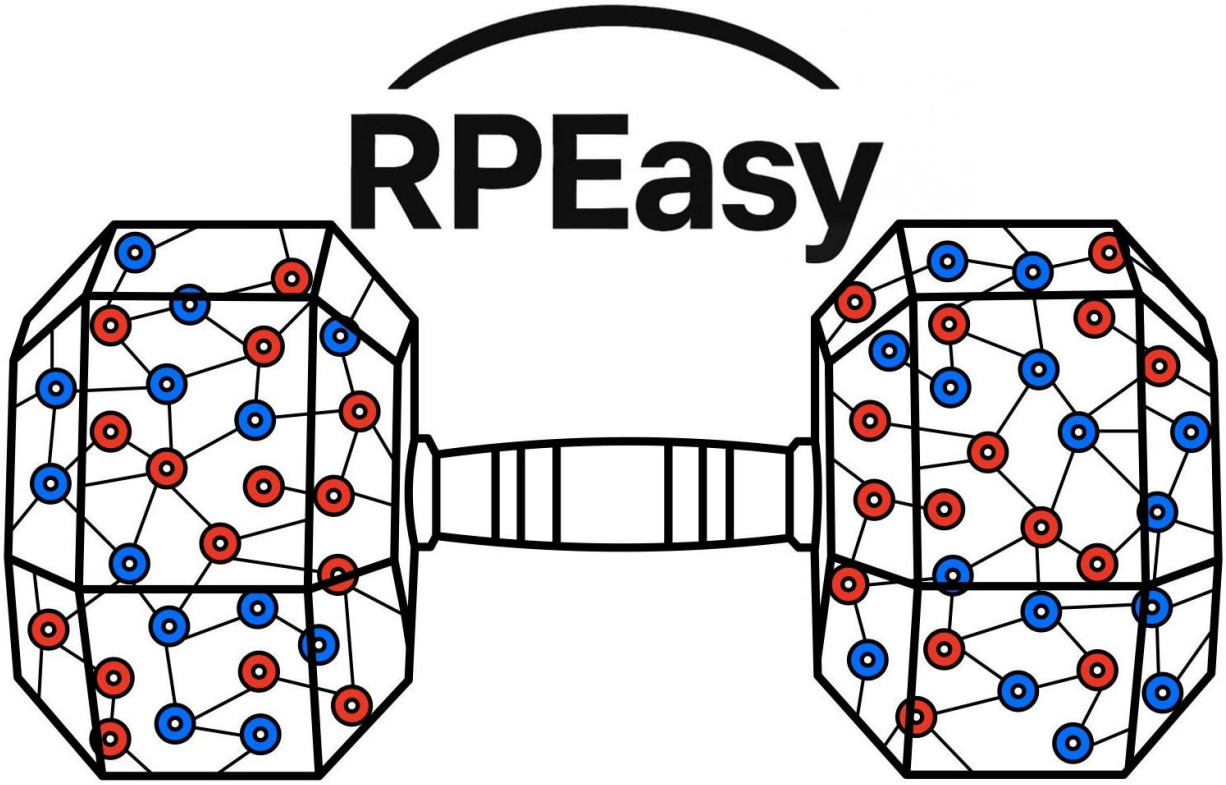
ELEC 490/498 Progress Report



Project title: RPEasy

Group #: 53

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Progress So far

Dataset

The current dataset consists of 92 videos (32 bench press, 25 deadlifts, and 35 squat) which are all collected from the internet. These where originally 56 videos before data augmentation techniques were used to trim videos and modify the length of a set. The RPE for all 92 videos were labeled by Trevor Felix, a certified level 5 personal trainer at GoodLife Fitness.

Facial Strain Analysis

The development of the OpenFace feature extractor was completed on schedule and followed the blueprint correctly with some fine-tuning still to be conducted to improve performance. The OpenFace model was tested first on initial facial tracking, then on landmark placement accuracy, and finally on Facial Action Unit (AU) performance. The current pipeline crops the video onto the lifters face, extracts all OpenFace features, isolates key features that correlate most to physical exertion, and then finally chooses only the max observed values (crucial in determining RPE) of the features for model training. The primarily obstacles were having the model properly identify the lifters face in the video and identifying the features that correlate best with RPE (which could still be subject to some changes).

Bar Speed Tracking

A hybrid computer vision model was implemented to extract the bar speed change from the users squat, deadlift, and bench press. The system utilizes MMPose to initially detect the wrist coordinates of the lifter, then utilizes OpenCV to monitor vertical displacement frame-by-frame. We engineered custom functionality that filters out static pauses and automatically segments the movement into the concentric (lifting) and eccentric (lowering) phases. By using the standard diameter of a weight plate (45cm) as a spatial reference, the system converts pixel movement into meters per second (m/s). This allows us to extract high value features such as first rep velocity, last rep velocity, and velocity loss (Δv), which directly correlate with neuromuscular fatigue.

Body Posture

Initial setup of body posture analysis was completed on schedule, with ongoing fine-tuning of the model to increase accuracy and consistency. The MMPose model has challenges with accurate tracking at certain recording angles or when parts of the body are obscured, such as by a squat rack or part of the weights. By enforcing rules about where the model is allowed to place key points within a video, the model can better track the joints of a lifter by taking inter-frame motion and partial obscuration into account. Once joint key points have been placed on a video, the stream of data must be mapped onto a more condensed metric to be fed into the LGBM regressor. The method chosen for this is to compare the change in joint positioning between the first and last rep of a set. This condenses the high-dimensional frame-by-frame key point coordinates into a single scalar value to be used by the LGBM regressor.

LGBM Regressor

We have successfully implemented a functional LGBM regressor that predicts RPE by integrating bar speed change with facial muscle activation features (Action Units). Due to the constraints of our small dataset (n=92), a 5-fold cross-validation was used to ensure the model's reliability rather

than relying on a single train/test split. Currently, the model results in a Mean Absolute Error (MAE) of 0.83, however the model is currently underfitting due to the small dataset. By integrating body posture features and expanding the dataset, we expect to resolve the current underfitting and reduce the LGBM regressor's error to our ±0.5 MVP target.

Changes Made

Computer Vision Models

The blueprint originally mentioned three key computer vison models used in extracting important physical exertion features: OpenFace for facial strain, MMPose for pose estimation, and MMDetection for bar speed tracking.

After attempting to implement MMDetection for bar speed, we experienced dependency issues and unstable tracking, causing us to shift to the more stable and consistent OpenCV model for tracking. MMDetection proved to be computationally heavy and unnecessary for velocity tracking. It struggled to maintain a consistent ID for the barbell across frames as it would lose the bar when it passed by a dark shirt or the rack. B Soy switching to OpenCV, we treat the task as motion tracking rather than object detection. We now initialize a tracking point on the wrist or plate once and follow its vector using optical flow. This approach is significantly faster, computationally light, and produces smoother velocity curves with fewer tracking errors.

MediaPipe was incorporated into the OpenFace pipeline to improve face tracking in the videos. We noticed the facial strain model was having difficulty finding the face due to background noise and other people found in the videos. To combat this, we implemented MediaPipe to first crop the video onto the primary centered person visible in the frame, and then further isolate onto the face so OpenFace has a clearer and more stable image of the weightlifter.

Data Collection

Another change straying from the blueprint was the amount and source of data collected for the project. Initially, the dataset was intended to consist primarily of recorded weightlifting sets collected by the group with supplementary data collected online from public spaces where required. With our current progress, the balance of the dataset has shifted and now is predominantly data collected online with our own recorded sets being added overtime.

This change occurred because collecting data from online platforms proved to be a more time-efficient and easier method. Online sources like YouTube, TikTok, and Instagram contain an abundance of readily available videos from weightlifters, making it much more practical. Although we are recording more sets of our group members now, the online data will likely remain the primary source.

This change could have been anticipated as recording sets can be time-consuming and requires consistent personal commitment and discipline, which is sometimes disrupted by day-to-day responsibilities, weekly schedules, and life events.

Plan Going Forward

Dataset

Over the course of the project, team members have been expanding the dataset by recording their own lifts. Additionally, the dataset has been expanding with new videos from the internet. Currently the dataset is at 356 videos compared to the 92 that the current LGBM regressor is using. The team plans to have a total of 500 videos, labeled by Trevor, to complete the dataset.

Training and Testing Pipeline

Now that all the models are set up, they will all be combined as one functioning AI pipeline. The training pipeline will take all 500 labeled videos and run them through the video preprocessor (crop videos to focus on main lifter). It will then feed these 500 videos into the 3 computer vision models which will then output the results into a master spread sheet. This master spread sheet will be used to train the LGBM regressor model. The testing pipeline will follow the same layout as the training pipeline but will now only take 1 video that the user uploads to find their predicted RPE. This video will run through the preprocessor and the 3 computer vision models. The extracted features will be fed into the LGBM regressor to predict the RPE of the set.

Regarding the Gantt chart, it states that the back-end/front-end of the pipeline will finish development in week 14/15 but currently we are on track to finish on week 15/16. This is due to the computer vision models taking longer to finalize then anticipated. However, the Gantt chart schedule is aggressive as it was intended to provide the team with more time to tune the final model. The team still has sufficient time to tune to model as there will still be 9 weeks remaining after the back-end pipeline is finalized.

Model Tuning

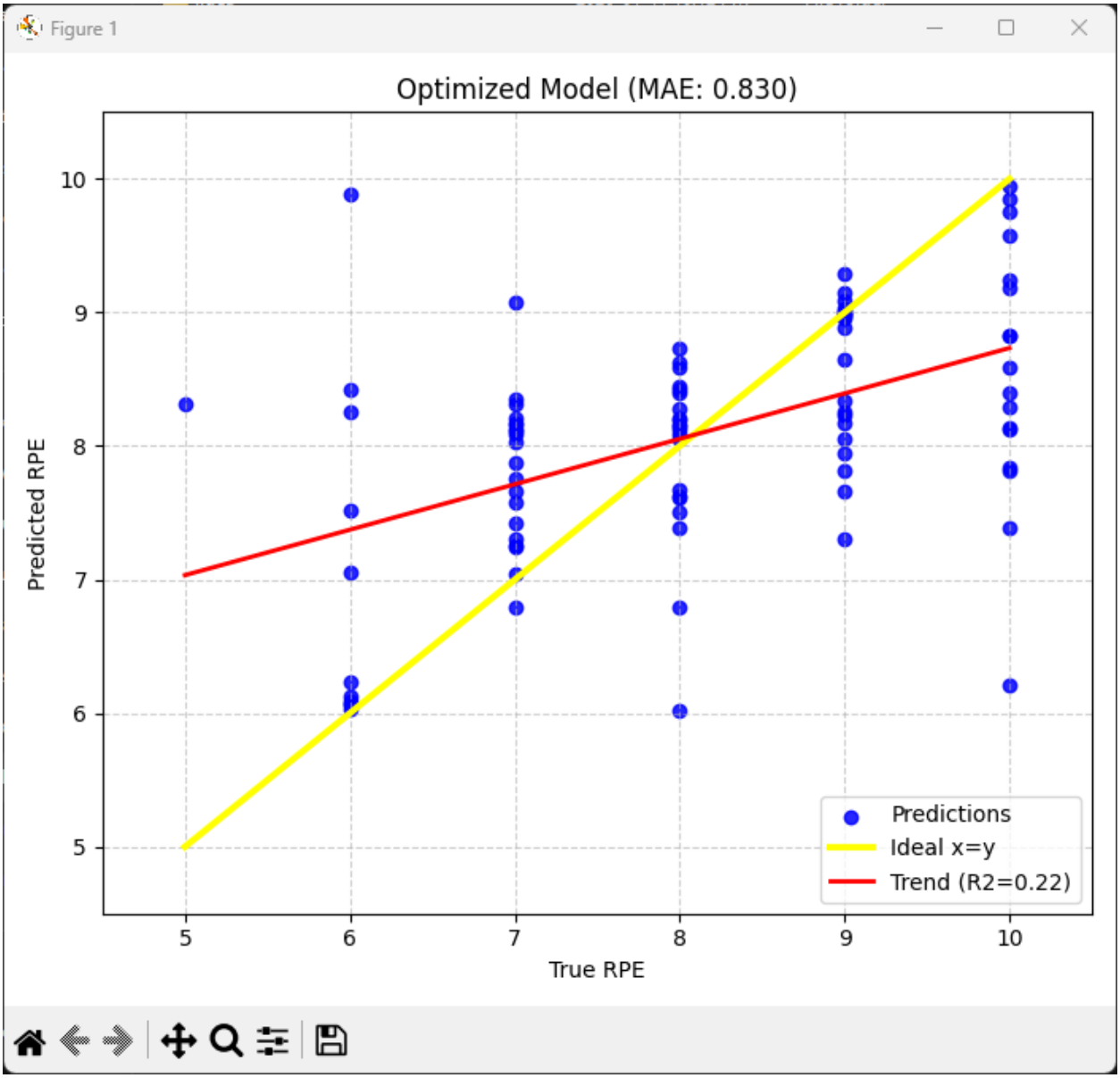
Each individual computer vision model, as well as the LGBM regressor, will undergo iterative fine tuning to improve accuracy metrics. This includes improving computer vision tracking and LGBM hyperparameters using the current dataset, as well as improving computer vision models to deal with issues that may arise from new videos being added to the dataset. A significant part of this tuning process will be using cross-validation to test different methods of interpreting computer vision data before they are fed into the regressor, such as analysing different facial action units and different functions for calculating change in form with MMPose.

UI/App Design

The final stage of the project involves developing a user interface to make the RPE estimation accessible to end-users. Currently, our AI models operate as standalone scripts; the next milestone is to integrate them into a cohesive backend framework (likely using Flask or FastAPI) that can serve predictions to a frontend application. We plan to develop a mobile-responsive web application where users can upload a video of their set, which is then processed asynchronously by our pipeline. The UI will focus on simplicity, displaying the predicted RPE alongside a breakdown of the contributing factors (e.g., "High Facial Strain detected" or "Significant Velocity Loss detected") to provide actionable feedback to the athlete.

Appendix

LGBM Regressor



	A	B	C	G	J	M	P	S	V	Y	AB	AE	AH	AK
1	video_name	True_RPE	delta_speed_m_s	confidence_mean	AU04_max	AU06_max	AU07_max	AU09_max	AU10_max	AU12_max	AU17_max	AU20_max	AU25_max	AU26_max
2	Bench Press 1.mp4	7	0.286966063	0.907511839	1.9525	1.0375	2.15	0.655	1.8475	1.11	2.185	1.825	2.0925	2.0525
3	Bench Press 2.mp4	10	0.284162411	0.851725664	2.16	0.55	0.78	0.9	1.24	0.55	2.59	1.42	2.61	1.92
4	Bench Press 3.mp4	8	1.248880875	0.970714286	1.41	1.09	2.38	3.33	1.1	1.21	3.33	1.18	1.91	3.66
5	Bench Press 4.mp4	9	1.358036328	0.896011323	1.815	1.475	1.41	0.75	0.95	1.56	2.615	3.57	2.425	1.99
6	Bench Press 5.mp4	10	1.446974136	0.849024457	1.726666667	0.623333333	1.63	0.8	1.23333333	1.07333333	2.56666667	1.47333333	2.27	1.72
7	Bench Press 6.mp4	7	0.203976211	0.907511839	1.9525	1.0375	2.15	0.655	1.8475	1.11	2.185	1.825	2.0925	2.0525
8	Bench Press 7.mp4	8	0.267489278	0.913237559	1.42	0.756666667	1.49666667	1.68666667	0.95333333	0.61	2.44333333	1.08	2.41666667	2.23
9	Bench Press 8.mp4	9	0.453911812	0.896011323	1.815	1.475	1.41	0.75	0.95	1.56	2.615	3.57	2.425	1.99
10	Bench Press 9.mov	7	0.347975345	0.783333333	2.13	0.72	1.56	0.67	2.15	0.75	5	3.09	3.22	3.16
11	Bench Press 10.mp4	10	0.335137694	0.830277778	0.84	0.49	1.88	0.66	0.78	1.26	2.52	0.66	1.33	1.02
12	Bench Press 11.mp4	10	0.307264904	0.849024457	1.726666667	0.623333333	1.63	0.8	1.23333333	1.07333333	2.56666667	1.47333333	2.27	1.72
13	Bench Press 12.mp4	9	0.313720299	0.896011323	1.815	1.475	1.41	0.75	0.95	1.56	2.615	3.57	2.425	1.99
14	Bench Press 13.mp4	6	1.622056314	0.873443633	2.146666667	0.836666667	2.585	1.3	2.62333333	0.92833333	3.985	2.34166667	2.47333333	3.11
15	Bench Press 14.mp4	9	0.313720299	0.896011323	1.815	1.475	1.41	0.75	0.95	1.56	2.615	3.57	2.425	1.99
16	Bench Press 15.mp4	7	0.343310185	0.88617284	2.94	2.32	1.49	0.54	1.38	2.79	1.38	1.49	1.94	1.88
17	Bench Press 16.mp4	9	0.420096167	0.890298507	2.94	2.32	1.49	0.69	1.38	2.79	2.16	3.81	2.2	2.62
18	Bench Press 17.mp4	6	0.502817886	0.873443633	2.146666667	0.836666667	2.585	1.3	2.62333333	0.92833333	3.985	2.34166667	2.47333333	3.11
19	Bench Press 18.mp4	8	0.502764526	0.913237559	1.42	0.756666667	1.49666667	1.68666667	0.95333333	0.61	2.44333333	1.08	2.41666667	2.23
20	Bench Press 19.mp4	9	0.313720299	0.896011323	1.815	1.475	1.41	0.75	0.95	1.56	2.615	3.57	2.425	1.99
21	Bench Press 20.mp4	8	0.182827687	0.913237559	1.42	0.756666667	1.49666667	1.68666667	0.95333333	0.61	2.44333333	1.08	2.41666667	2.23
22	Bench Press 21.mp4	7	0.062867265	0.849170984	2.16	0.55	0.78	0.84	1.24	0.55	2.62	1.51	2.58	1.96
23	Bench Press 22.mp4	8	0.202409264	0.851725664	2.16	0.55	0.78	0.9	1.24	0.55	2.59	1.42	2.61	1.92
24	Bench Press 23.mp4	10	0.339498112	0.86506993	2.18	0.83	2.23	0.84	1.68	1.41	2.59	2.34	2.87	2.22
25	Bench Press 24.mp4	7	1.836945009	0.940526316	0.69	0.63	1.33	0.09	0.52	0	1.35	0.7	1.09	1.14
26	Bench Press 25.mp4	8	0.641336578	0.917272727	0.69	0.63	1.33	0.83	0.52	0.07	1.41	0.64	2.73	1.11
27	Bench Press 26.mp4	9	0.711590321	0.901724138	0.69	0.63	1.33	0.81	0.52	0.33	3.07	3.33	2.65	1.36
28	Bench Press 27.mp4	9	0.129997068	0.896011323	1.815	1.475	1.41	0.75	0.95	1.56	2.615	3.57	2.425	1.99
29	Bench Press 28.mp4	10	0.129766665	0.849024457	1.726666667	0.623333333	1.63	0.8	1.23333333	1.07333333	2.56666667	1.47333333	2.27	1.72
30	Bench Press 29.mp4	7	0.374649506	0.954177215	2.02	0.65	5	1.15	4.25	1.1	3.39	3.6	2.76	3.23
31	Bench Press 30.mp4	5	0.301929733	0.970714286	1.41	1.09	2.38	3.33	1.1	1.21	3.33	1.18	1.91	3.66
32	Bench Press 31.mp4	6	0.257063792	0.873443633	2.146666667	0.836666667	2.585	1.3	2.62333333	0.92833333	3.985	2.34166667	2.47333333	3.11
33	Bench Press 32.mp4	6	0.311721026	0.9125	2.29	1.48	5	2.76	4.27	2.64	5	5	1.83	4.6
34	Squat 1.mp4	6	0.157062245	0.855711493	2.313333333	0.653333333	1.24333333	0.93333333	1.96	0.48333333	3.52666667	1.61	2.60333333	2.44
35	Squat 2.mp4	10	-0.199122727	0.873489347	2.078	0.911	2.363	0.978	1.484	0.932	3.248	2.418	2.665	2.638
36	Squat 3.mp4	8	0.344643885	0.86925816	2.68	0.78	2.29	0.79	2.12	1.56	4.47	2.95	2.57	3.29
37	Squat 4.mp4	8	-0.241925635	0.87627482	2.244	0.693	1.87	1.61	1.949	0.976	3.865	3.37	3.584	3.104

OpenFace

```python
class OpenFaceExtractor:

    # Key Action Units for physical exertion
    EXERTION_AUS = [
        'AU04_r',  # Brow Lowerer (concentration/strain)
        'AU06_r',  # Cheek Raiser (squinting/strain)
        'AU07_r',  # Lid Tightener (strain)
        'AU09_r',  # Nose Wrinkler (strain)
        'AU10_r',  # Upper Lip Raiser (strain)
        'AU12_r',  # Lip Corner Puller (grimace)
        'AU17_r',  # Chin Raiser (strain)
        'AU20_r',  # Lip Stretcher (strain)
        'AU25_r',  # Lips Part (breathing/exertion)
        'AU26_r',  # Jaw Drop (breathing/exertion)
    ]

    # All AU intensity columns
    ALL_AU_INTENSITY = [
        'AU01_r', 'AU02_r', 'AU04_r', 'AU05_r', 'AU06_r', 'AU07_r',
        'AU09_r', 'AU10_r', 'AU12_r', 'AU14_r', 'AU15_r', 'AU17_r',
        'AU20_r', 'AU23_r', 'AU25_r', 'AU26_r', 'AU45_r'
    ]

    # AU classification columns (presence/absence)
    ALL_AU_CLASSIFICATION = [
        'AU01_c', 'AU02_c', 'AU04_c', 'AU05_c', 'AU06_c', 'AU07_c',
        'AU09_c', 'AU10_c', 'AU12_c', 'AU14_c', 'AU15_c', 'AU17_c',
        'AU20_c', 'AU23_c', 'AU25_c', 'AU26_c', 'AU28_c', 'AU45_c'
    ]

    def __init__(self, verbose=True):

        self.verbose = verbose
        self._check_openface()

    def _check_openface(self):
        """Verify OpenFace binary is available."""
        if not ofu.check_openface_binary():
            raise RuntimeError("OpenFace binary not found. Cannot extract features.")
```

```python
class PoseGuidedFaceDetector:
    def __init__(self, pose_model=None, person_detector=None, verbose=True):

        self.verbose = verbose
        self._load_mediapipe()

    def _load_mediapipe(self):

        try:
            import mediapipe as mp
            self._mp_pose = mp.solutions.pose.Pose(
                static_image_mode=False,
                model_complexity=1,
                enable_segmentation=False,
                smooth_landmarks=True,
                min_detection_confidence=0.5,
                min_tracking_confidence=0.5
            )
            if self.verbose:
                print("✓ Using MediaPipe Pose for face region detection")
                print("  (Video mode with temporal smoothing enabled)")
        except ImportError:
            raise ImportError("MediaPipe required. Install with: pip install mediapipe")

    def get_head_bbox_from_pose(self, keypoints, img_shape, confidence_threshold=0.3):
        """
        Calculate head bounding box from body keypoints.
```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	video_nam	error	detection_	detection_	total_frame	confidenc	confidenc	confidenc	AU04_max	AU04_q95	AU04_peal	AU06_max	AU06_q95	AU06_peal	AU07_max	AU07_q95	AU07_peal	AU09_max	AU09_q95	AU09_peal	AU10_max	AU10_q95	AU10_peal
2	Bench Press 1.mp4		0.442733	93	219	0.907512	0.073278	0.625	1.9525	1.2585	10.405	1.0375	0.72875	2.077612	2.15	1.227625	2.15	0.655	0.493375	0.655	1.8475	0.8525	1.8475
3	Bench Press 10.mp4		0.270677	36	133	0.830278	0.141512	0.03	0.84	0.7915	16.8	0.49	0.171	0.49	1.88	1.699	1.88	0.66	0.24	0.66	0.78	0.162	0.78
4	Bench Press 11.mp4		0.755731	182.6667	215.3333	0.849024	0.099272	0.523333	1.726667	1.186167	7.046667	0.623333	0.262667	0.623333	1.63	1.074833	1.63	0.8	0.412167	0.8	1.233333	0.608167	1.233333
5	Bench Press 12.mp4		0.294873	81.5	279	0.896011	0.08628	0.4	1.815	1.4085	12.80263	1.475	1.20175	5.425132	1.41	1.035	1.41	0.75	0.4775	0.75	0.95	0.6485	0.95
6	Bench Press 13.mp4		0.365242	70.33333	244.3333	0.873444	0.155356	0.153333	2.146667	1.596667	3.595854	0.836667	0.482917	0.836667	2.585	1.79425	7.486111	1.3	0.752083	1.3	2.623333	1.782917	3.740662
7	Bench Press 14.mp4		0.294873	81.5	279	0.896011	0.08628	0.4	1.815	1.4085	12.80263	1.475	1.20175	5.425132	1.41	1.035	1.41	0.75	0.4775	0.75	0.95	0.6485	0.95
8	Bench Press 15.mp4		0.409091	81	198	0.886173	0.10954	0.03	2.94	2.243	36.75	2.32	2.133	6.480447	1.49	1.182	1.49	0.54	0.32	0.54	1.38	1.048	1.38
9	Bench Press 16.mp4		0.487273	134	275	0.890299	0.089105	0.03	2.94	2.183	24.91525	2.32	2.0365	10.22026	1.49	1.063	1.49	0.69	0.39	0.69	1.38	0.823	1.38
10	Bench Press 17.mp4		0.365242	70.33333	244.3333	0.873444	0.155356	0.153333	2.146667	1.596667	3.595854	0.836667	0.482917	0.836667	2.585	1.79425	7.486111	1.3	0.752083	1.3	2.623333	1.782917	3.740662
11	Bench Press 18.mp4		0.449796	92	184	0.913238	0.062635	0.786667	1.42	1.017833	1.42	0.756667	0.501833	0.756667	1.496667	1.036167	1.496667	1.686667	0.468333	1.686667	0.953333	0.760667	0.953333
12	Bench Press 19.mp4		0.294873	81.5	279	0.896011	0.08628	0.4	1.815	1.4085	12.80263	1.475	1.20175	5.425132	1.41	1.035	1.41	0.75	0.4775	0.75	0.95	0.6485	0.95
13	Bench Press 2.mp4		1	226	226	0.851726	0.07781	0.77	2.16	1.3785	2.16	0.55	0.087	0.55	0.78	0.0755	0.78	0.9	0.511	0.9	1.24	0.76	1.24
14	Bench Press 20.mp4		0.449796	92	184	0.913238	0.062635	0.786667	1.42	1.017833	1.42	0.756667	0.501833	0.756667	1.496667	1.036167	1.496667	1.686667	0.468333	1.686667	0.953333	0.760667	0.953333
15	Bench Press 21.mp4		1	193	193	0.849171	0.079552	0.77	2.16	1.289	2.16	0.55	0.079	0.55	0.78	0.08	0.78	0.84	0.508	0.84	1.24	0.8	1.24
16	Bench Press 22.mp4		1	226	226	0.851726	0.07781	0.77	2.16	1.3785	2.16	0.55	0.087	0.55	0.78	0.0755	0.78	0.9	0.511	0.9	1.24	0.76	1.24
17	Bench Press 23.mp4		0.996516	286	287	0.86507	0.078493	0.77	2.18	1.3885	2.18	0.83	0.53	0.83	2.23	1.45	2.23	0.84	0.4855	0.84	1.68	0.9025	1.68
18	Bench Press 24.mp4		0.118012	19	161	0.940526	0.048816	0.88	0.69	0.65	0.69	0.63	0.422	0.63	1.33	1.098	1.33	0.09	0.082	0.09	0.52	0.488	0.52
19	Bench Press 25.mp4		0.103774	22	212	0.917273	0.075035	0.77	0.69	0.645	0.69	0.63	0.396	0.63	1.33	1.069	1.33	0.83	0.524	0.83	0.52	0.484	0.52
20	Bench Press 26.mp4		0.102473	29	283	0.901724	0.083455	0.77	0.69	0.634	0.69	0.63	0.367	0.63	1.33	1.007	1.33	0.81	0.565	0.81	0.52	0.474	0.52
21	Bench Press 27.mp4		0.294873	81.5	279	0.896011	0.08628	0.4	1.815	1.4085	12.80263	1.475	1.20175	5.425132	1.41	1.035	1.41	0.75	0.4775	0.75	0.95	0.6485	0.95
22	Bench Press 28.mp4		0.755731	182.6667	215.3333	0.849024	0.099272	0.523333	1.726667	1.186167	7.046667	0.623333	0.262667	0.623333	1.63	1.074833	1.63	0.8	0.412167	0.8	1.233333	0.608167	1.233333
23	Bench Press 29.mp4		0.243827	79	324	0.954177	0.055206	0.82	2.02	0.852	2.02	0.65	0.281	0.65	5	2.5505	5	1.15	1.0635	1.15	4.25	1.074	4.25
24	Bench Press 3.mp4		0.245614	28	114	0.970714	0.035059	0.82	1.41	1.03	1.41	1.09	1.0225	1.09	2.38	1.964	2.38	3.33	0.37	3.33	1.1	1.038	1.1
25	Bench Press 30.mp4		0.245614	28	114	0.970714	0.035059	0.82	1.41	1.03	1.41	1.09	1.0225	1.09	2.38	1.964	2.38	3.33	0.37	3.33	1.1	1.038	1.1
26	Bench Press 31.mp4		0.365242	70.33333	244.3333	0.873444	0.155356	0.153333	2.146667	1.596667	3.595854	0.836667	0.482917	0.836667	2.585	1.79425	7.486111	1.3	0.752083	1.3	2.623333	1.782917	3.740662
27	Bench Press 32.mp4		0.254682	68	267	0.9125	0.143624	0.03	2.29	1.857	2.29	1.48	0.7495	1.48	5	3.33	5	2.76	1.0145	2.76	4.27	2.7255	4.27
28	Bench Press 4.mp4		0.294873	81.5	279	0.896011	0.08628	0.4	1.815	1.4085	12.80263	1.475	1.20175	5.425132	1.41	1.035	1.41	0.75	0.4775	0.75	0.95	0.6485	0.95
29	Bench Press 5.mp4		0.755731	182.6667	215.3333	0.849024	0.099272	0.523333	1.726667	1.186167	7.046667	0.623333	0.262667	0.623333	1.63	1.074833	1.63	0.8	0.412167	0.8	1.233333	0.608167	1.233333
30	Bench Press 6.mp4		0.442733	93	219	0.907512	0.073278	0.625	1.9525	1.2585	10.405	1.0375	0.72875	2.077612	2.15	1.227625	2.15	0.655	0.493375	0.655	1.8475	0.8525	1.8475
31	Bench Press 7.mp4		0.449796	92	184	0.913238	0.062635	0.786667	1.42	1.017833	1.42	0.756667	0.501833	0.756667	1.496667	1.036167	1.496667	1.686667	0.468333	1.686667	0.953333	0.760667	0.953333

Bar Speed

video_name	rep_count	first_rep_speed_m_s	last_rep_speed_m_s	delta_speed_m_s
Bench Press 1.mp4	3	0.438363262	0.151397199	-0.286966063
Bench Press 2.mp4	3	0.438363262	0.154200851	-0.284162411
Bench Press 3.mp4	4	1.375574007	0.126693133	-1.248880875
Bench Press 4.mp4	5	1.528415564	0.170379235	-1.358036328
Bench Press 5.mp4	6	1.528415564	0.081441428	-1.446974136
Bench Press 6.mp4	4	0.331840542	0.127864331	-0.203976211
Bench Press 7.mp4	5	0.339557764	0.072068487	-0.267489278
Bench Press 8.mp4	4	0.53337083	0.079459018	-0.453911812
Bench Press 9.mov	4	0.563598945	0.215623599	-0.347975345
Bench Press 10.mp4	1	0.276377755	0.276377755	0
Bench Press 11.mp4	2	0.405105343	0.097840438	-0.307264904
Bench Press 12.mp4	1	0.514361145	0.514361145	0
Bench Press 13.mp4	7	1.813500789	0.191444475	-1.622056314
Bench Press 14.mp4	1	0.171344918	0.171344918	0
Bench Press 15.mp4	5	0.528139309	0.184829125	-0.343310185
Bench Press 16.mp4	6	0.515257863	0.095161696	-0.420096167
Bench Press 17.mp4	3	0.697504896	0.19468701	-0.502817886
Bench Press 18.mp4	3	0.680825283	0.178060756	-0.502764526
Bench Press 19.mp4	1	0.558377773	0.558377773	0
Bench Press 20.mp4	5	0.330062909	0.147235223	-0.182827687
Bench Press 21.mp4	13	0.691292388	0.628425124	-0.062867265
Bench Press 22.mp4	14	0.66745472	0.465045455	-0.202409264
Bench Press 23.mp4	15	0.645206229	0.305708117	-0.339498112
Bench Press 24.mp4	2	2.023572115	0.186627106	-1.836945009
Bench Press 25.mp4	3	2.023572115	1.382235536	-0.641336578
Bench Press 26.mp4	4	2.023572115	1.311981794	-0.711590321
Bench Press 27.mp4	3	0.232536654	0.102539586	-0.129997068
Bench Press 28.mp4	4	0.232536654	0.102769988	-0.129766665
Bench Press 29.mp4	10	0.533938603	0.159289097	-0.374649506
Bench Press 30.mp4	3	0.524333021	0.222403288	-0.301929733
Bench Press 31.mp4	4	0.535070431	0.278006638	-0.257063792
Bench Press 32.mp4	10	0.672303861	0.360582835	-0.311721026
Squat 1.mp4	3	0.333332704	0.176270459	-0.157062245
Squat 2.mp4	2	0.355001505	0.554124233	0.199122727
Squat 3.mp4	1	0.341812187	0.341812187	0
Squat 4.mp4	4	0.570990799	0.812916434	0.241925635
Squat 5.mp4	4	0.440022055	0.084860618	-0.355161437
Squat 6.mp4	5	0.440022055	0.119026019	-0.320996036
Squat 7.mp4	4	0.440773884	0.176375824	-0.264398061
Squat 8.mp4	5	0.440773884	0.162402143	-0.278371742
Squat 9.mp4	4	0.385382647	0.192442357	-0.19294029
Squat 10.mp4	5	0.385382647	0.217883463	-0.167499184
Squat 11.mp4	4	0.335354153	0.193828027	-0.141526127
Squat 12.mp4	5	0.335354153	0.185876097	-0.149478056
Squat 13.mp4	4	0.486711608	0.208500417	-0.278211191
Squat 14.mp4	6	0.486711608	1.199082573	0.712370965
Squat 15.mp4	4	0.263194258	0.172145649	-0.09104861
Squat 16.mp4	5	0.255883307	0.161614184	-0.094269123
Squat 17.mp4	4	0.526130518	0.282050757	-0.244079761
Squat 18.mp4	5	0.526130518	0.141236804	-0.384893714
Squat 19.mp4	5	0.526130518	0.051496857	-0.474633661
Squat 20.mp4	4	0.38673706	0.201488906	-0.185248154
Squat 21.mp4	1	0.236373947	0.236373947	0
Squat 22.mp4	4	0.44140122	0.221791379	-0.219609841
Squat 23.mp4	5	0.459057268	0.079330342	-0.379726926
Squat 24.mp4	4	0.453011225	0.192183801	-0.260827424
Squat 25.mp4	5	0.435587716	0.198010052	-0.237577664
Squat 26.mp4	4	0.34050815	0.16721943	-0.173288721
Squat 27.mp4	6	0.34050815	0.287153067	-0.053355083
Squat 28.mp4	5	0.385081518	0.096279204	-0.288802315
Squat 29.mp4	2	0.309064258	0.14956924	-0.159495018
Squat 30.mp4	5	0.518514469	0.156839015	-0.361675454
Squat 31.mp4	6	0.577042229	0.300417836	-0.276624393
Squat 32.mp4	7	0.577042229	0.270573916	-0.306468313
Squat 33.mp4	8	0.577042229	0.241660138	-0.335382091
Squat 34.mp4	9	0.577042229	0.219929248	-0.357112981
Squat 35.mp4	10	0.577042229	0.154072011	-0.422970218
Deadlift 1.mp4	1	0.042744302	0.042744302	0
Deadlift 2.mp4	2	0.075582406	0.081345978	0.005763572
Deadlift 3.mp4	1	0.15879265	0.15879265	0
Deadlift 4.mp4	1	0.029019206	0.029019206	0
Deadlift 5.mp4	1	0.06072596	0.06072596	0
Deadlift 6.mp4	2	0.071581124	0.034584368	-0.036996755
Deadlift 7.mp4	2	0.01695541	0.13944097	0.12248556
Deadlift 8.mp4	1	0.019996392	0.019996392	0
Deadlift 9.mp4	1	0.128570315	0.128570315	0
Deadlift 10.mp4	1	0.044596899	0.044596899	0
Deadlift 11.mp4	2	0.071581124	0.034584368	-0.036996755
Deadlift 12.mp4	1	0.039613572	0.039613572	0
Deadlift 13.mp4	4	0.228573925	0.200607556	-0.02796637
Deadlift 14.mp4	5	0.270178113	0.263274877	-0.006903237
Deadlift 15.mp4	5	0.05942413	0.110795513	0.051371383
Deadlift 16.mp4	5	0.270178113	0.263274877	-0.006903237
Deadlift 17.mp4	3	0.074489532	0.137994861	0.063505329
Deadlift 18.mp4	4	0.074489532	0.096500517	0.022010986
Deadlift 19.mp4	3	0.019480905	0.042674518	0.023193613
Deadlift 20.mp4	4	0.019480905	0.015788329	-0.003692576
Deadlift 21.mp4	1	0.081882138	0.081882138	0
Deadlift 22.mp4	2	0.08130922	0.062724764	-0.018584455
Deadlift 23.mp4	1	0.058753798	0.058753798	0
Deadlift 24.mp4	1	0.022223895	0.022223895	0
Deadlift 25.mp4	3	0.086243012	0.057687877	-0.028555135

MMPose

```
DATA FIELDS
keypoints: array([[[ 549.375,  631.875],
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                   [ 530.625,  631.875],
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                   [ 643.125,  706.875],
                   [ 474.375,  725.625],
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                   [ 736.875,  706.875],
                   [ 399.375,  725.625],
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                   [ 530.625,  950.625],
                   [ 680.625, 1138.125],
                   [ 474.375, 1138.125],
                   [ 661.875, 1363.125],
                   [ 474.375, 1344.375]]])
bboxes: array([[   0.,    0., 1080., 1920.]], dtype=float32)
bbox_scores: array([1.], dtype=float32)
keypoint_scores: array([[0.4472444 , 0.4815387 , 0.42997319, 0.475867  , 0.2808933 ,
                   0.51910967, 0.40314525, 0.6568272 , 0.5668707 , 0.63464546,
                   0.613933  , 0.47179103, 0.471623  , 0.49294686, 0.6563078 ,
                   0.44475538, 0.6609423 ]], dtype=float32)
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